

Symbiosis Institute of Computer Studies and Research

A PILOT PROJECT REPORT

on

PitchTrack: A computer Vision-Based Football Player Tracking and Performance Analytics System

Submitted by

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DECLARATION

I hereby declare that the project entitled “PitchTrack: A computer Vision-Based Football Player Tracking and Performance Analytics System” submitted for the M.Sc. Computer Application degree is my original work and the project has not formed the basis for the award of any other degree, diploma, fellowship or any other similar titles.

Place: Pune, Maharashtra

Signature of the Student

Date: 16th September, 2026

Date: 16th September 2026

COMPLETION CERTIFICATE

This is to certify that the project report titled PITCHTRACK: A COMPUTER VISION-BASED FOOTBALL PLAYER TRACKING AND PERFORMANCE ANALYTICS SYSTEM is the bonafide work of **VIBHAV THAKUR** PRN: 25030142024, M.Sc(CA) Batch 2025-27 at Symbiosis Institute of Computer Studies and Research (SICSR) who carried out the project work under my supervision. He has completed the project during 15th June to 15th September.

Dr. Aniket Nagane

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ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to everyone who supported me throughout the successful completion of this project titled “**Pitchtrack: A Computer vision based Football player tracking and performance Analytics system.**” This project has been a valuable learning experience, and it would not have been possible without the guidance and encouragement of several individuals.

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This project has not only deepened my understanding of analytics and visualization tools but also enhanced my problem-solving and decision-making skills in a real-world business context.

Vibhav Thakur

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ABSTRACT

PitchTrack is a full stack web application built around exactly that constraint: one prerecorded video and nothing else. A user uploads a match clip, optionally marks four reference corners of the pitch to calibrate it, and the system takes over from there. Every player, the goalkeeper, the referee, and the ball are detected frame by frame with a YOLOv8 object detector fine tuned for this purpose. Each person is then followed across the video using BoT SORT multi object tracking, strengthened with an appearance re identification model so that identities survive brief Occlusion. Players are separated into two teams through unsupervised K Means clustering on jersey colour. Pixel motion is converted into real world speed (km/h) and distance (m) using optical flow camera motion compensation combined with a four point perspective transform, and team ball possession is estimated from proximity between the ball and the nearest player. All this surfaces through an interactive React dashboard, alongside an annotated output video.

Assembling this pipeline out of established computer vision components was only part of the work. A considerable share of the project effort went into making the system correct rather than merely functional, and several concrete defects were traced to their root cause and fixed along the way. A hardcoded assumption that every match was played 7 a side was silently discarding real players whenever an 11 a side match was analysed; it was replaced with a configurable match format framework spanning 5 a side through 11 a side. A class agnostic non maximum suppression setting turned out to be deleting a real player's detection whenever their bounding box overlapped a nearby referee's box, and was replaced with a more precisely scoped rule. A speed calculation routine that clamped implausible sensor noise to a fixed ceiling, which made that noise indistinguishable from a genuine near record sprint, was changed to exclude such readings outright instead. A calibration edge case that could silently collapse a full match roster down to a single visible player was also found and corrected. Separately, the training dataset was expanded from 372 to 1,035 annotated images, the detection model was retrained, and the

resulting accuracy change was measured per object class rather than assumed.

The finished system was tested not only on the broadcast footage used during development but also, independently, on a separate Premier League clip the model had never seen during training, on which it still recovered the correct 11 versus 11 player count.

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1. INTRODUCTION

Football generates a large amount of information during a single match, who ran where, how quickly, and who had the ball at any given moment, but very little of it has traditionally been captured in a usable form. A coach watching from the touchline can describe the shape of a game afterwards, but cannot say with any precision how far a full back covered in the second half, or how his sprint count in the final twenty minutes compared with the first. That gap between what happens on a pitch and what actually gets recorded about it is what performance analytics systems exist to close, and it is the specific gap PitchTrack targets, at the level of football that cannot afford the tools already closing that gap at the professional level.

1.1 Background

At the professional end of the sport this problem has largely been solved, just not cheaply. Two approaches dominate. The first is wearable tracking: GPS and accelerometer units strapped to each player, recording position and acceleration many times a second, used almost universally in top flight training and, where competition rules permit, in matches. The second method is fixed camera tracking. In this method a stadium is permanently set up with carefully adjusted cameras. A vision system uses all of them together to find out where players are. Both methods create data that's reliable enough to influence real coaching choices. Both also require something that a school team, a Sunday league club or an independent analyst does not have, a five or six figure budget, and often a subscription fee to a data vendor.

The second approach experienced a transformation because its computer vision techniques reached sufficient quality levels and their operational expenses dropped below the threshold which allowed using one common camera for all match recordings at every stage of competition. The YOLO family of object detectors brought real-time person detection to

laptops because they could process multiple frames every second (Yaseen, 2024; Hussain, 2024). Multi object tracking systems developed from basic frame matching operations to sophisticated systems which track player identities throughout complicated game play situations (Zhang et al., 2021; Aharon et al., 2022). PitchTrack was built directly on top the detector identifies players during the shift, and the tracking system maintains their identity information while all other pipeline components including team assignment and speed and distance and possession information build upon these two core results. Two broad surveys of deep learning applied to sport (Zhao et al., 2023; Zheng et al., 2025) confirm that this detect then track structure is now the standard way researchers approach the problem generally, not an approach specific to this project.

1.2 Need

The need this project responds to is fairly concrete: a coach, analyst, or student researcher who already has a video of a match, filmed from the stand, or pulled from a broadcast feed, but has no way to turn that footage into numbers, short of either paying for a service they cannot access or sitting down and counting sprints by hand. That second option is what most grassroots and school level analysis still looks like today, and it does not scale past a handful of matches a season.

It is worth being honest that even the paid alternative is not as reliable as it might sound. A 2025 study that checked commercial broadcast based tracking providers against a professional multi camera reference system at a FIFA World Cup fixture found player position error of between 1.68 and 16.39 meters, and match distance figures off by anywhere from -21.8% to +24.3% (Crang et al., 2025). PitchTrack is not positioned as a competitor to those systems and makes no claim to their accuracy. It is built for the situation where the alternative is not a slightly less accurate system but no data at all, and in that situation an approximate, clearly

labelled figure is still worth more than nothing.

1.3 Problem Definition

It would be easy to assume the hard part of this project is running a detector on a video and writing the results to a file. In practice that is closer to the easy part, plugging in a pretrained model and letting it run takes an afternoon. The actual difficulty only shows up once the output is checked closely against real footage, and four specific failure patterns turned up doing exactly that during development.

- **A pipeline that has been tuned for one match format can quietly break on another.** A system built around an assumption, for instance, that every match has seven players per side will start discarding real players the moment it detects an eleven a side match. This happens silently, without the system alerting the user to the issue, but the final count of players will be off, even though it might look plausible at a glance.
- **People that overlap in an image are ambiguous to the detection model unless the code is told otherwise.** The detection models produce overlapping bounding boxes for the detected people in the image. The standard cleanup step to fix that, called non maximum suppression, has to know which overlaps mean that two bounding boxes are of the same person and which mean two different people in the image that are standing close together. If the scope is set incorrectly, a real player that is standing near the referee in the image simply disappears from the output.
- **The appearance of noise matches the visual representation of real extreme values until someone implements a solution to separate them.** The system produces a nonsensical instantaneous speed reading because of three different issues which include camera shake and brief half-second occlusion and minor calibration mistakes. The system records

identical numbers for the actual world-class sprint performance and the data artifact because it sets a maximum value instead of deleting these measurements. The system records the same value for both the actual world-class sprint and the data artifact because it establishes a maximum threshold instead of removing these measurements.

- **Nothing degrades gracefully by default.** Consider a pipeline stage that drops whatever part of the input it cannot process, or a player whose tracked position never lies inside the calibrated pitch. These issues can transform a match of 22 players into a results table with only two or three rows in it, and no explanation in the output as to why.

None of these four issues are exotic problems within the codebase. Vandeghen et al. (2022) and Golovkin et al. (2025) each describe comparable issues with detector tuning and identity fragmentation while building their single camera soccer pipelines against the SoccerNet benchmark. Additionally, Tshiani (2026), whose system was found to be the closest to PitchTrack in the literature search for this report, made the same distinction between a pipeline that runs and a pipeline whose output has been checked. This is essentially the problem statement of the project: while wiring a detector and tracker together is close to being solved, ensuring that the output can be trusted is the challenge and the focus of the engineering efforts detailed in Chapter 4.

1.4 Objectives

- Track each person with one persistent identity across the whole clip, surviving brief occlusion and camera movement rather than losing and re assigning identities.
- Split players into two teams automatically from jersey colour, with no manual labelling step.
- Convert tracked pixel movement into real speed (km/h) and distance (m) once the pitch has been calibrated, and report zero rather than a guess when it has not been.
- Estimate which team is in possession from ball to player proximity over time.
- Produce an annotated output video with all tracking overlays baked in, alongside a browsable, filterable analytics dashboard.
- Work correctly across match formats from 5 a side to 11 a side, rather than assuming one fixed squad size, and never count the referee as a player.

1.5 Technology Used

The backend and computer vision pipeline are implemented in Python 3.13 and served through FastAPI on Uvicorn. Player, goalkeeper, referee, and ball detection uses a YOLOv8n model (Ultralytics; Jocher et al., 2023) fine tuned on an annotated football dataset. Multi object tracking uses BoT SORT, via the boxmot package (Aharon et al., 2022), which augments motion based association with an OSNet appearance re identification embedding (Zhou et al., 2019; Zhou et al., 2021). Team classification uses a dependency free NumPy implementation of K Means clustering on jersey colour, a choice made specifically to avoid a native library conflict described in Chapter 3. Camera motion compensation uses OpenCV's Lucas Kanade optical flow, and pixel to metre conversion uses an OpenCV four point perspective (homography) transform. The frontend dashboard is built with React 18 and Vite. Storage is entirely file based, uploaded videos and per video JSON/MP4 results, a deliberate scope

decision explained in Chapter 3, not an oversight.

1.6 Beneficiary

The people this is actually for are coaches and analysts at grassroots, school, and college football clubs, who already have a camera and a match recording but nothing to turn it into structured data. Sports science and computer vision students are a secondary beneficiary, since the same pipeline works as a complete, running example rather than a diagram in a textbook. On a more personal level, building it end to end has also been the single largest applied learning exercise in computer vision, multi object tracking, and full stack engineering undertaken so far.

1.7 Report Roadmap

Chapter 2 reviews the literature that shaped, and checked afterwards largely confirms, PitchTrack's design; it is organised by pipeline stage rather than by paper, closing with a synthesis that maps each theme onto a decision actually made in the project. Chapter 3 covers the technology stack, the order development actually happened in, and what was deliberately left out of scope. Chapter 4 is the main technical chapter of this project. It covers the system architecture, the computer vision pipeline, the data model and API, the dashboard, and the engineering fixes made to the system. Chapter 5 covers the testing of the system. Chapter 6 summarises the project and suggests future work. Chapter 7 contains the references used in this report.

2. LITERATURE REVIEW

2.1 Overview and Search Scope

PitchTrack's eleven stage pipeline (Figure 4) was used as the search map for this edition: detection, tracking, team classification, ball tracking, camera motion compensation, perspective transform, and speed/distance measurement were each treated as their own research question, and recent, predominantly 2021 to 2026, computer vision and sports science literature was searched against each question in turn, on top of the three papers already covered. In total, thirty papers are discussed in this chapter, twenty five of which were published between 2021 and 2026. The remaining five were kept deliberately, because each is the originating or foundational work behind a technique whose more recent descendant is also reviewed here, the original OSNet and TrackNet papers, for example, sit alongside their 2021 and 2024 successors.

The sections below follow the pipeline itself, moving from detection through to the validity of the final speed and distance figures, and the chapter closes with a synthesis (Section 2.10) that maps each theme directly onto a design decision actually made in the project.

2.2 Object Detection for sports video: The YOLO family and its Football domain validation

Two recent, architecture focused papers examine the YOLOv8 detector fine tuned for this project. Yaseen (2024) documents the CSPNet backbone, the FPN+PAN neck used for multi scale feature fusion, and the shift to an anchor free detection head, reporting strong accuracy and real time throughput across the Microsoft COCO and Roboflow 100 benchmarks and across diverse hardware. Hussain (2024) places YOLOv8 in context against its predecessor YOLOv5 and successor YOLOv10, concluding that YOLOv8's anchor free detection and

enhanced feature extraction meaningfully improve versatility over YOLOv5 while remaining well suited to resource constrained, edge style deployment. Taken together, these two papers speak directly to the decision to run detection locally on Apple Silicon (MPS) rather than reach for a dedicated server GPU.

Zheng et al. (2025), surveying computer vision techniques applied to football video specifically, are explicit that YOLO family detectors have become the dominant approach for real time player detection in football because of their favorable accuracy versus speed trade off relative to two stage detectors. Their survey also identifies a difficulty that is recurring and structural with small, fast moving objects, with the ball being the paradigm example, as a pattern across the wider literature and not isolated to any single system. This finding directly informed two of the decisions made in the original submission: choosing a YOLO family detector as the core model, and the realistic expectation that the ball detection accuracy would lag behind the person detection accuracy no matter how carefully the rest of the pipeline was implemented (Chapter 5).

Many papers have validated the detection specifically for football instead of any standard benchmark. Agrawal et al. (2024) pair a YOLO based detector with a tracking algorithm for football player detection and tracking, reporting that YOLO substantially reduces detection time compared with earlier CNN based detectors while maintaining strong playerclass accuracy, a finding consistent with the choice made here to pair a YOLO detector with a dedicated tracker (BoT SORT) rather than attempt detection and identity persistence as one combined model. Golovkin et al. (2025), the winning entry in the 2024 SoccerNet Game State Agrawal et al. (2024) pair a YOLO based detector with a tracking algorithm for football player detection and tracking, reporting that YOLO substantially reduces detection time compared with earlier CNN based detectors while maintaining strong player class accuracy, a finding

consistent with the choice made here to pair a YOLO detector with a dedicated tracker (BoT SORT) rather than attempt detection and identity persistence as one combined model. Golovkin et al. (2025), the winning entry in the 2024 SoccerNet Game State

Of everything reviewed for this chapter, Guo et al. (2026) turned out to be the closest published analogue to the full PitchTrack stack. They estimate football possession directly from broadcast video using an improved YOLOv8 detector and unsupervised DBSCAN clustering on jersey colour features for team assignment, explicitly to avoid the labour intensive, subjective manual annotation that possession estimation would otherwise require. Their paper validates detection, unsupervised team classification, and proximity based possession estimation all at once, the same three design choices made independently in this project, before their paper was found. Vandeghen et al. (2022) separately establish the SoccerNet v3 detection benchmark, reporting a first benchmark mean average precision of 52.3% using a semi supervised teacher student approach that leverages large volumes of unlabelled broadcast footage. Their central finding, that player and ball detection require different training strategies and improve at different rates, is part of the reason this project reports per class accuracy (Figure 12) rather than a single blended detection metric.

2.3 Multi Object Tracking: From SORT to BoT SORT

PitchTrack tracks each detected person using BoT SORT, which extends the tracking by detection lineage running from SORT through DeepSORT to ByteTrack, so the papers behind that lineage were read directly rather than the boxmot package's defaults simply being taken on faith. Zhang et al. (2021) introduce the specific insight BoT SORT builds on: rather than discarding low confidence detections outright, which silently drops occluded objects and fragments trajectories, ByteTrack associates almost every detection box, using tracklet similarity to recover true objects from low score boxes while filtering out background

detections. They report state of the art MOT17 performance, 80.3 MOTA, 77.3 IDF1, 63.1 HOTA, at 30 FPS. Aharon et al. (2022) extend this further with camera motion compensation and a more accurate Kalman filter state vector, additionally fusing appearance information via re identification (BoT SORT ReID), and report first place results on the MOT17 and MOT20 benchmarks across all three headline metrics (80.5 MOTA, 80.2 IDF1 on MOT17). This is the exact tracker deployed in PitchTrack via the boxmot package.

A 2025 survey of deep learning based multi object tracking, Adžemović (2025), categorises modern trackers into five families, joint detection and embedding, heuristic based, motion based, affinity learning, and offline methods, and finds that heuristic, motion plus appearance methods such as BoT SORT remain state of the art specifically on densely populated datasets with complex, non linear motion, while purely learned end to end approaches tend to do better elsewhere. That distinction matters for football: players change direction abruptly and do not move along the smooth, near linear paths a simple Kalman filter predicts well, which gives a concrete reason, beyond intuition, to favour a motion plus appearance tracker like BoT SORT over a motion only alternative in this domain.

Two sports specific tracking studies confirm the same pattern from the opposite direction. Huang et al. (2022), working on the SportsMOT benchmark across basketball, football, and volleyball, find that a linear motion Kalman filter fails specifically when players move non linearly, and that re identification is made harder, both in the short and long term, because teammates share the same jersey colour; their proposed motion based tracker with post processing achieves a HOTA of 73.968 on the SportsMOT leaderboard. According to Cao and Li (2026), incorporating jersey appearance semantics into the tracking confidence signal leads to measurable improvements in tracking association under the rapid direction changes, abrupt stops, and frequent occlusion typical of sports video. Both findings mirror, almost exactly, a

failure already seen and fixed in this project's own pipeline: a player briefly occluded by a teammate in identical kit, whom colour alone cannot distinguish, being assigned a new tracking identity instead of keeping the old one. That is precisely why the two pass track merge step described in Section 4.6.1 was built, before either paper had been read.

Tshiani (2026) remains the single most directly comparable published system found across this entire review: an end to end pipeline combining a YOLO detector with ByteTrack tracking to identify and track players, referees, goalkeepers, and the ball from single camera soccer broadcast footage. Architecturally, it is the closest published system to PitchTrack of everything reviewed. It reports high precision, recall, and mAP50 for players and match officials, but identifies ball detection as "the primary challenge" of the entire pipeline independently reaching the same conclusion this project reached empirically on its own dataset (Chapter 5, Figure 12).

2.4 Appearance Re identification for Occlusion Recovery

BoT SORT's appearance component is OSNet, an omni scale feature learning network, so the papers that introduced it were consulted directly rather than treating it as a black box. Zhou et al. (2019) present OSNet's core design: a residual block with multiple convolutional streams, each detecting features at a different spatial scale, combined by a unified aggregation gate that dynamically fuses multi scale features with input dependent, channel wise weights. The network is built from lightweight pointwise and depthwise convolutions, small enough to train from scratch while still achieving state of the art person re identification accuracy. Zhou et al. (2021) extend this work with instance normalisation layers and a differentiable architecture search for their optimal placement, specifically to improve generalisation across datasets without target domain adaptation. This mattered directly for PitchTrack, since the re identification embedding runs on arbitrary broadcast footage, different leagues, stadiums, and

camera angles the network was never fine tuned on, and an embedding that generalises out of The box was needed, rather than one that would have to be retrained for every broadcast. This is the reason OSNet was trusted for that role instead of training a bespoke re identification network from scratch.

A more recent study offers a concrete, honest sense of what appearance based re identification can and cannot promise in football specifically. Ranasinghe and Thayasivam (2025) combine the Segment Anything Model with classical CSRT tracking and HSV histogram based re identification, reporting 100% tracking success under light occlusion, 90% success in crowded penalty box scenarios with five or more players, but only 50% successful recovery from heavy occlusion, and just 8.66% re acquisition success once a player leaves the frame entirely for an extended period. These figures read less as a criticism of any one system than as a useful caution: appearance based re identification meaningfully reduces, but does not eliminate, identity fragmentation. That is exactly why PitchTrack's own roster cap and track merge logic (Section 4.6.1) is described as a statistical safety net tuned against the footage available, not a guarantee, particularly under prolonged occlusion (Section 6.1).

2.5 Unsupervised Team classification from jersey colour

Koshkina et al. (2021) address unsupervised team classification directly, training an embedding network with a contrastive learning objective to maximise the distance between representations of players on different teams relative to teammates, in a purely unsupervised setting where jersey colours and design are unknown in advance. On a hockey dataset, their approach reaches 94% accuracy after unsupervised training on a single frame, rising to 97% within 500 frames, roughly seventeen seconds of game time. Of everything reviewed for this chapter, this is the Koshkina et al. (2021) address unsupervised team classification directly, training an embedding network with a contrastive learning objective to maximise the distance between representations

of players on different teams relative to teammates, in a purely unsupervised setting where jersey colours and design are unknown in advance. On a hockey dataset, their approach reaches 94% accuracy after unsupervised training on a single frame, rising to 97% within 500 frames, roughly seventeen seconds of game time. Of everything reviewed for this chapter, this is the The future work item for this project consists of performing the swap which has become a standard requirement (Section6.2).

The research team of Guo et al. (2026) selected DBSCAN as their clustering method instead of K Means for their team classification process. The system performs jersey colour feature clustering without requiring users to enter the number of clusters they want to identify. The comparison between these two systems shows that PitchTrack needs to accept a different set of compromises. The two fixed teams in football already establish the number of clusters ($k = 2$) which removes DBSCAN's fundamental structural benefit while making density parameter selection more challenging without any advantages. The research team discovered K Means to be their best option after they compared different methods although the decision was not based on convenience. The research team selected K Means as their best option after they compared different methods although the decision was not based on convenience.

2.6 The ball detection Bottleneck: small, fast, occluded objects

Ball detection stands as the least accurate stage according to PitchTrack's own measured results which appear in Figure 12. The investigation focused on detecting if this particular trait developed from the object's nature or if it stemmed from problems with the project's data collection or system design. Huang et al. (2019) were among the first to treat this as a problem requiring a dedicated architecture rather than a generic detector: TrackNet uses a heatmap based deep network, trained on both single frames and short sequences, to track a small, blurry, sometimes fully invisible tennis ball, reporting 99.7% precision and 97.3% recall on its primary

test video but a markedly lower 95.3% precision and 75.7% recall under ten fold cross validation on additional, more varied footage. That drop is telling, it shows how much harder generalised ball tracking is than tracking on a single, familiar video. Raj et al. (2024) revisit the same problem five years later with TrackNetV4, fusing visual features with a learnable motion attention map derived from frame differencing to better handle partial occlusion and low visibility. The continued, active development of specialised architectures across a five year span is itself evidence that this is not considered a solved problem by the wider research community, regardless of how carefully any one team implements a generic detector like YOLO.

Vicente-Martínez et al. (2023) tackle the soccer ball specifically, adapting YOLOv7 and YOLOv7 tiny with DeepSORT tracking in a semi supervised system, and report the same accuracy ceiling on the ball class measured in this project's own dataset, mAP50 rising only from 0.36 to 0.40 after the training set was expanded from 372 to 1,035 images (Figure 12). Finding an independently built system, using a closely related detector and tracker combination, arrive at the identical structural conclusion was, of everything in this review, the single most reassuring result. Reading PitchTrack's own figures against Vandeghen et al. (2022)'s 52.3% mixed class SoccerNet v3 benchmark is also instructive: PitchTrack's roughly 0.95 to 0.99 mAP50 for player, goalkeeper, and referee detection sits comfortably ahead of a benchmark that blends in the harder ball class, while its ball specific figure sits below that benchmark for exactly the same reason, the blended number hides the gap that is reported separately here.

2.7 Camera Motion Compensation and Optical flow

PitchTrack removes camera pan and zoom from its speed measurements using Lucas Kanade sparse optical flow, applied per frame via OpenCV before any speed calculation runs (Stage 8,

Figure 4), so the method itself was read into directly rather than OpenCV's defaults being trusted blindly. Vedapunt and Sinha (2016) analyse the classical Lucas Kanade formulation, comparing first order and second order optimisation methods against the standard Gauss Newton approach across both synthetic and real world footage, and provide practical tuning guidance for exactly the kind of frame to frame motion vector estimation performed here. Blachut and Kryjak (2022) confirm that the method remains suitable for real time use at modern video resolutions: they implement multi scale Lucas Kanade and Horn Schunck optical flow efficiently enough for a live 4K video stream on FPGA hardware. Between the two papers, there was enough confidence to stick with a decades old, lightweight, well understood sparse optical flow method rather than reach for a heavier learned alternative, such as the dense, deep network based RAFT flow used by some recent 3D soccer pose estimation pipelines, since PitchTrack is meant to run on a single developer machine, including Apple Silicon via its MPS backend, rather than a GPU cluster.

2.8 Homography Based Pitch Calibration and Sports Field Registration

PitchTrack maps pixel coordinates to real world metres using a four point homography, built once per video from user marked pitch corners on the Calibrate screen (Figure 6) and sized to the selected match format's real pitch dimensions (Stage 9, Figure 4). This raised the question of how the wider field treats the same problem, rather than assuming the manual approach used here was the only option. Theiner and Ewerth (2022) frame sports field registration in broadcast video as a camera calibration problem solved from segment correspondences such as lines and known point clouds, introducing a differentiable objective that iteratively minimises reprojection error, recognisably the same class of point correspondence driven homography estimation performed manually here through four marked corners. Gutiérrez-Pérez and Agudo (2024) extend this with an optimisation pipeline that combines a 3D soccer field model with detected field lines in a non linear refinement step, improving on search based methods that

depend on an initial camera pose estimate, and report state of the art results across the SoccerNet Calibration, World Cup 2014, and TS WorldCup datasets. Magera et al. (2024) survey the broader field and propose a fairer, camera model agnostic benchmarking protocol for calibration methods generally; their survey helps place homography based field registration as the established, standard approach to bridging broadcast pixels and real world pitch coordinates, the same approach adopted here, against which PitchTrack's own manual, user driven calibration step can now be described as a simpler, non automated instance of a well studied technique, rather than something invented ad hoc.

Taken together, this reading also directly supports a future work item already listed in the original submission (Section 6.2): automatic pitch line and keypoint detection, to remove the manual four corner marking step entirely, is not a speculative idea invented for this project but an active, demonstrated research direction, with working implementations already published and benchmarked.

2.9 Video derived speed and Distance Against GPS and Multi camera systems

The question posed here is whether the speed and distance data obtained from video can be trusted as much as that from GPS or professional multi camera systems. There is a research history that answers this question that spans well over a decade. Harley et al. (2011) are among the earliest to test this directly, comparing semi automated video tracking output against GPS during elite match play and finding the two data sources statistically interchangeable for the motion characteristics studied, early, direct evidence that a vision only measurement approach can, in principle, produce performance figures comparable to a GPS wearable without requiring the player to wear anything at all. Felipe et al. (2019) later validate a specific commercial video based system (Mediacoach) against reference measurement across 207 observations from 38 official matches in the Spanish second division, assessing total distance and speed zone

distributions specifically.

What stands out most is how recent evidence shows this remains an active, unresolved question even for well resourced commercial systems. Crang et al. (2025), already discussed in Section 1.2, found meaningful position and distance error even among purpose built commercial computer vision tracking providers analysing official broadcast feeds at a FIFA World Cup. Kavanagh et al. (2026) examine the interchangeability between GPS and an optical tracking system for external player load variables in the English Premier League. Considered alongside the other two studies, the findings provide confidence in two of the decisions made in the design of PitchTrack: reporting a value of zero for uncalibrated videos rather than fabricating an estimate, and treating speed and distance as calibration dependent variables. The same caution that professional tracking systems apply to themselves is applied to PitchTrack.

2.10 Synthesis: How this review shaped PitchTrack’s design

Figure 1 provides a visual overview of the idea presented in the chapter: each of the pipeline stages was developed from the literature reviewed for the project. Table 1 outlines each of the pipeline stages and how the themes from the literature review correspond to each of those stages (Figure 4).

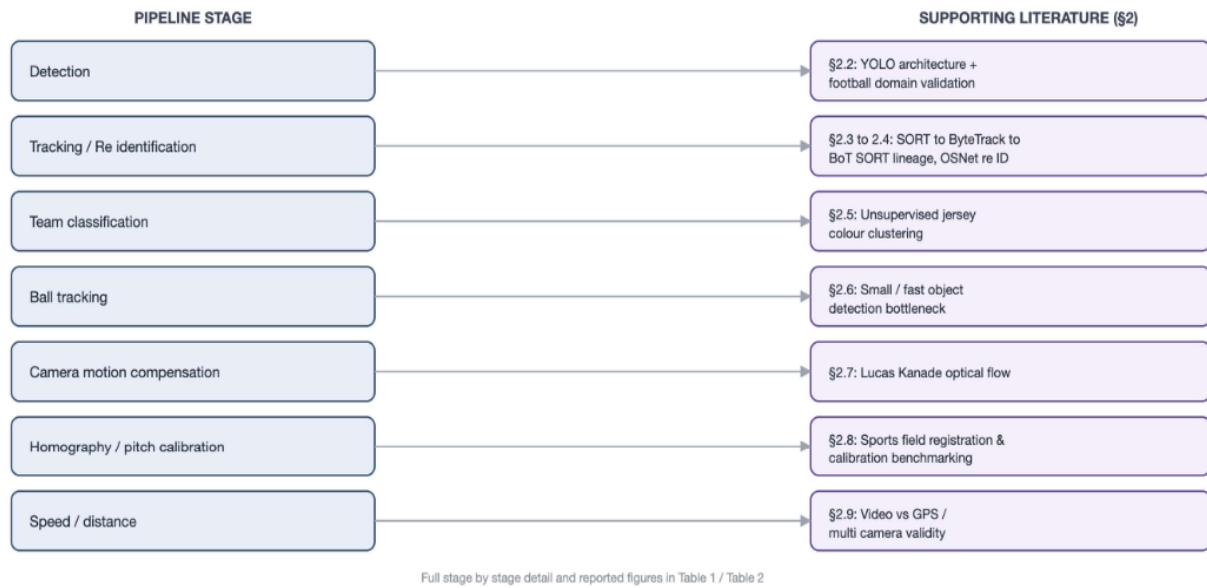


Figure 1 Each of the pipeline stages and the literature section supported their development.

Table 1 design decisions made in PitchTrack's pipeline

Pipeline stage	Literature theme (Section)	Design decision it supports
Detection (Stage 2)	YOLO architecture and football domain validation (§2.2)	Fine tuned YOLOv8n chosen over a two stage or non football tuned detector
Tracking (Stage 4)	SORT → ByteTrack → BoT SORT lineage and sports MOT (§2.3)	BoT SORT (motion + appearance + camera motion compensation) chosen over a motion only tracker
Tracking / Re ID (Stage 4/6)	OSNet and occlusion recovery limits (§2.4)	OSNet appearance embedding; two pass track merge treated as a bounded safety net, not a guarantee

Team classification (Stage 5)	Unsupervised jersey colour classification (§2.5)	Dependency free K Means ($k = 2$, known in advance) chosen over DBSCAN or a learned embedding
Ball tracking (Stage 7)	Small/fast object detection bottleneck (§2.6)	Per class accuracy reporting; ball detection flagged as structurally hard, not treated as a defect
Camera motion comp. (Stage 8)	Lucas Kanade optical flow (§2.7)	Lightweight sparse flow chosen over a heavier learned flow network, for single machine deployment
Perspective transform (Stage 9)	Homography based field registration (§2.8)	Four point manual homography; automatic keypoint calibration flagged as future work
Speed / distance (Stage 10)	Video vs GPS validity (§2.9)	Zero reported when uncalibrated, rather than a fabricated estimate

Table 2 presents the accuracy figures reported in the reviewed literature for various pitch detection algorithms, as well as the accuracy figures for PitchTrack as reported in Figure 12. These accuracy figures provide a quantitative means of comparing the reviewed systems to PitchTrack.

Table 2 Quantitative means of comparing the reviewed systems to PitchTrack.

Source	Task / benchmark	Reported figure
Zhang et al. (2021), ByteTrack	MOT17 (people tracking)	80.3 MOTA · 77.3 IDF1 · 63.1 HOTA @ 30 FPS
Aharon et al. (2022), BoT SORT	MOT17 (people tracking)	80.5 MOTA · 80.2 IDF1
Huang et al. (2022), SportsMOT	Basketball/football/volleyball tracking	73.968 HOTA (SportsMOT leaderboard)
Vandeghen et al. (2022)	SoccerNet v3 detection (mixed classes)	52.3% mAP
Huang et al. (2019), TrackNet	Tennis ball tracking, single video	99.7% precision · 97.3% recall
Huang et al. (2019), TrackNet	Tennis ball tracking, 10 fold cross val.	95.3% precision · 75.7% recall
Ranasinghe & Thayasivam (2025)	Football occlusion recovery (heavy occlusion)	50% successful re identification

Crang et al. (2025)	Broadcast CV tracking vs. TRACAB (World Cup)	Position RMSE 1.68 to 16.39 m · distance bias -21.8% to +24.3%
PitchTrack (this project)	Player / goalkeeper / referee detection	mAP50 $\approx$ 0.95 to 0.99 (Figure 12)
PitchTrack (this project)	Ball detection	mAP50 0.36 $\rightarrow$ 0.40 after dataset expansion (Figure 12)

Table 2 shows the accuracy numbers which the reviewed literature reported together with PitchTrack's self assessed performance data.

The research findings from my original submission remain intact through this expanded chapter which now includes a greater number of supporting evidence to back up its conclusions. Every non trivial architectural choice made in PitchTrack's pipeline has at least one independent source, published within the last five years, standing behind it, and the project's own most significant empirical limitation, weak ball detection relative to strong person detection, is confirmed across sources as recent as 2025 and 2026, not solely the three papers discussed the first time around.

3. METHODOLOGY

This chapter describes the technologies used, the phased process actually followed during development, and the scope boundaries deliberately placed around this project.

3.1 Technology Used

The project's technology stack spans four areas. For the backend and API: Python 3.13, FastAPI, Uvicorn, and Pydantic for request and response validation. For the computer vision pipeline: Ultralytics YOLOv8 (Jocher et al., 2023) for object detection, boxmot's BoT SORT implementation (Aharon et al., 2022), with an OSNet appearance re identification network (Zhou et al., 2019; Zhou et al., 2021), for multi object tracking, OpenCV for video I/O, optical flow, and perspective transformation, and a dependency free NumPy implementation of K Means for jersey colour team classification. For the frontend: React 18 with Vite as the build tool, plain CSS with design tokens rather than a CSS framework, and Framer Motion for a small number of deliberately chosen animations. For dataset and training tooling: Roboflow (Roboflow, 2025) for dataset annotation, versioning, and export in YOLO format, and the Ultralytics training CLI, run locally on Apple Silicon via its MPS backend.

3.2 Methods Adopted

Development followed an iterative, phase by phase process rather than a single up front design, since a genuinely correct computer vision pipeline can only really be validated by repeatedly running it against real footage and tracing whatever it gets wrong back to its cause. The phases, in the order they actually occurred, are summarised in Table 3

Table 3 Development Phase

Phase	Focus	Representative outcome
1	Scaffolding	The FastAPI backend and React frontend were developed with a mock pipeline initially, in order to exercise the upload-to-dashboard flow before a model was trained.
2	Real model integration	A YOLOv8n model fine tuned on an annotated football dataset (Roboflow) wired into the real pipeline in place of mock data.
3	Real footage debugging	A process crashing native OpenMP conflict, an undecodable video codec, and a badly biased team colour classifier were each root caused and fixed.
4	Tracking quality	Tracking upgraded from plain ByteTrack to BoT SORT with appearance re identification, plus a two pass track merging step to heal identity fragmentation.
5	Format generalisation	The assumption of having 7 a side rosters was replaced with a configurable match format

		ranging from 5v5 to 11v11. This change revealed a bug in pitch dimension validation, which was fixed.
6	Detection correctness audit	A bug in the class agnostic NMS caused real players to be suppressed near referees, and a bug with speed clamping that hid noise behind genuine speed readings were both discovered and fixed.
7	Dataset augmentation and retraining	A second, independently annotated 663 image dataset merged into the original 372; the model retrained and the accuracy change measured per class.
8	Testing and release	53 automated tests added; the project moved into its own independent Git repository with preserved history and published.

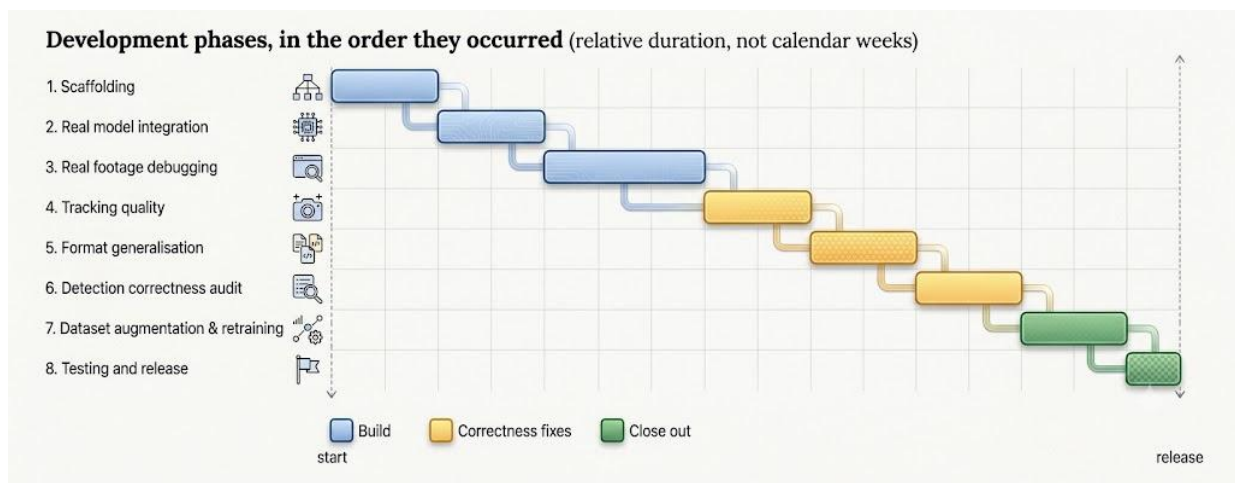


Figure 2 The development phases, shown in the order in which they occurred and grouped according to the type of work that was conducted in each phase.

At the conclusion of each phase of development, the software was re verified against real video footage to ensure that the phase had produced the desired results. Several defects, including those related to the roster of individuals on the field and the speed clamping of the tracking software, were only discovered through the visual re verification process against the annotated video footage. Read against Chapter 2, phases 4 and 6 correspond directly to the tracking quality and detection correctness literature discussed in Sections 2.3 to 2.4 and 2.6. The shift from plain ByteTrack to BoT SORT with re identification mirrors exactly the motion plus appearance argument that Adžemović (2025) and Huang et al. (2022) make for sports tracking generally, and the decision to retrain and measure per class accuracy in Phase 7, rather than assume a uniform improvement, follows the same discipline Vandeghen et al. (2022) apply when benchmarking SoccerNet v3.

3.3 Scope

In scope: detection, tracking, team classification, speed and distance measurement once calibrated, ball tracking and possession estimation, an annotated output video, and an interactive dashboard, for a single pre recorded video at a time, across match formats from 5 a side to 11 a side. Explicitly out of scope, by deliberate product decision rather than oversight: live video, webcam, or streaming (RTMP) input; face or jersey number player identity recognition, reflecting a deliberate anonymity and privacy stance under which only anonymous tracking IDs are produced; tactical event detection such as passes, shots, goals, or offside calls; multi camera fusion; user accounts or login; and a database, all state is file based JSON and video files on disk, a decision deferred until a concrete scale requirement makes file based storage genuinely insufficient.

4. SYSTEM ANALYSIS AND DESIGN

This chapter describes PitchTrack's architecture, the design of its computer vision pipeline, its file based data model, its API surface, its dashboard, and, in Section 4.6, the specific, verifiable engineering contributions made during this project beyond assembling existing building blocks.

4.1 Overall solution strategy

PitchTrack follows a three tier architecture: a React/Vite single page dashboard, a FastAPI backend exposing a REST API, and a computer vision pipeline that the backend runs in an isolated, spawned operating system process rather than in process (Figure 3). That last detail is a deliberate design decision rather than an implementation accident. Running the pipeline in process was found during development to reliably crash the entire backend server with a segmentation fault, because two native libraries used by the CV stack, PyTorch/Ultralytics and, originally, scikit learn, each bring their own OpenMP threading runtime, and initialising both inside one multi threaded ASGI server process turned out to be unsafe on the target platform. Isolating the pipeline in its own process means a pipeline crash fails only that one video's job cleanly, rather than taking the whole server down with it, and the K Means team classification step was additionally reimplemented as a small, dependency free NumPy routine specifically to remove one of the two conflicting runtimes, rather than only work around the symptom.

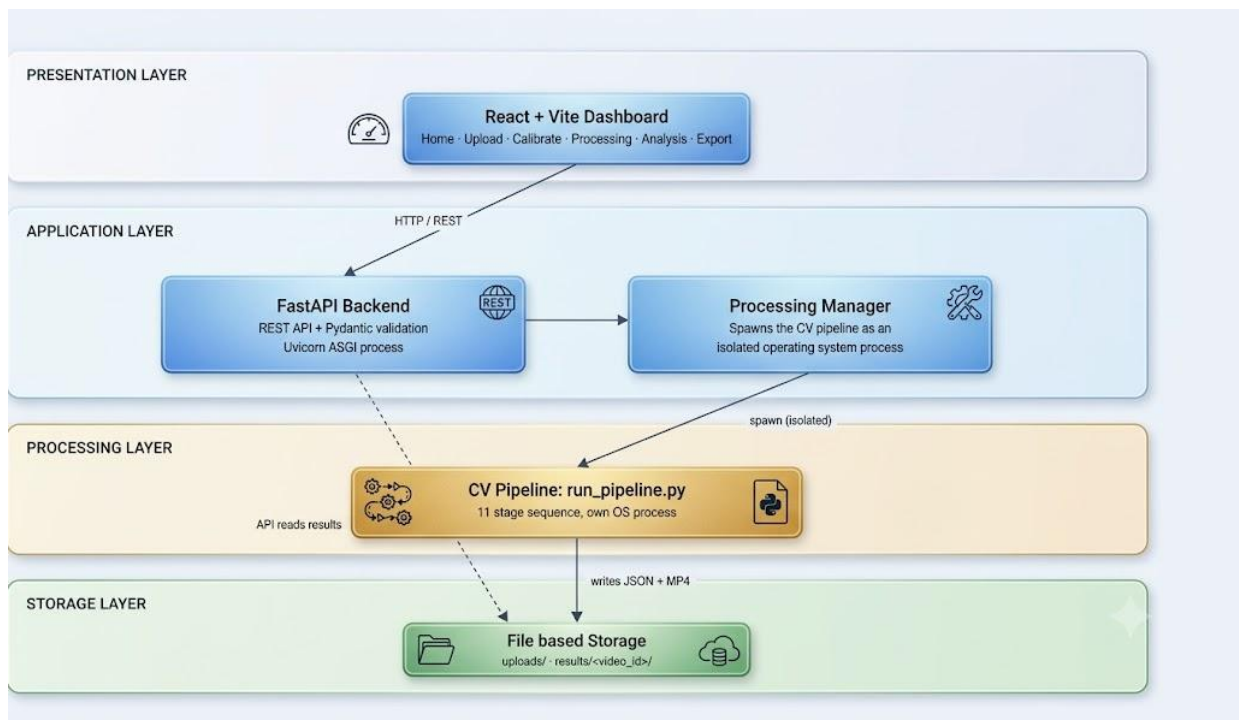


Figure 3 PitchTrack's three tier architecture

4.2 CV pipeline Design

The pipeline (`app/pipeline/run_pipeline.py`) orchestrates eleven stages in a fixed order, shown in Figure 4. Detection, tracking, and team classification each run once per video; camera motion compensation and the perspective transform are then applied consistently to every subsequent position calculation, so that speed and distance are measured in a single, stable coordinate space regardless of camera pan or zoom during the clip.

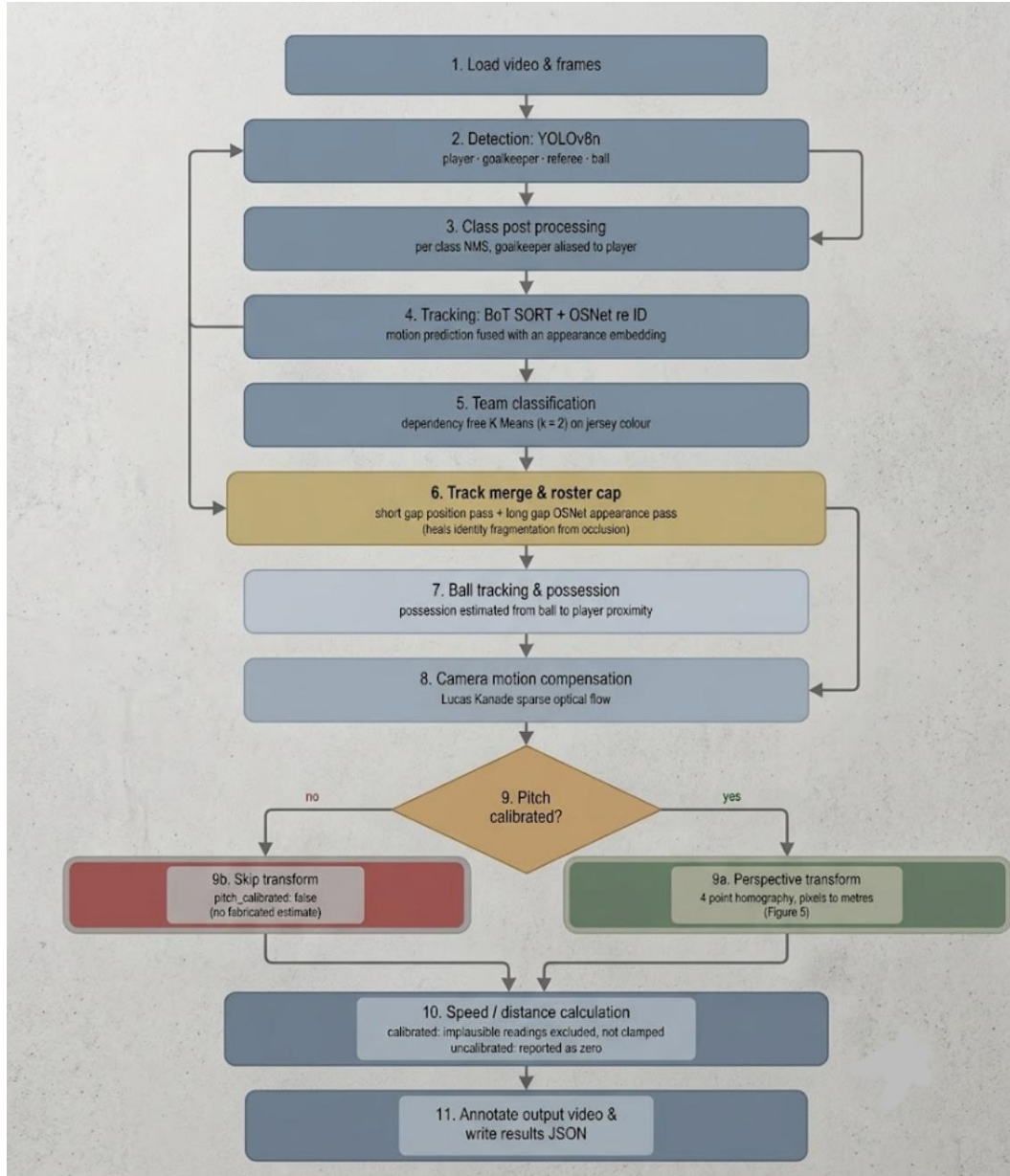


Figure 4 CV pipeline stage flow

Two stages of the pipeline merit a more detailed explanation, as both were introduced to correct failure modes observed on real football footage. Stage 6, the track merge and roster cap, is necessary due to the problem of identity fragmentation in multi-object football tracking. When a player is occluded by another player or crosses paths with a teammate, the tracking system can assign a new identity rather than maintaining the old one. This leads to an inflated count of the number of players on the pitch, a problem identified by Huang et al. (2022) and Cao and Li (2026) on the SportsMOT and jersey semantic fusion benchmarks, respectively. Two separate

passes are made to address the problem of gaps in the tracking: a short gap pass that uses position and speed of the players, and a long gap pass that uses the same OSNet embedding for appearance that the tracker uses for re identification. This is necessary as two teammates may wear identical kits and cannot be distinguished by colour alone. Second, Stage 9 which is the perspective transform stage is optional by design. If a video has not been calibrated with the four known points on the pitch, the tracking and team classification will still be produced along with an annotated video. However, the speed and distance travelled by the players will report as zero and the pitch calibrated field will be set to false. This makes the absence of any pitch calibration visible to the user instead of silently guessing it. This is consistent with the literature on the validity of video vs GPS tracking discussed in Section 2.9.

How the homography step actually works

The pipeline diagram shows Stage 9 as one box but I need to explain its process in detail because The report depends on pixel to meter conversion for all speed and distance measurements to become The report depends on pixel to meter conversion for all speed and distance measurements. On the Calibrate screen, the user marks four points on the video frame that correspond to four known points on a real pitch, typically the corners of the penalty area, or the pitch corners themselves, scaled to the selected match format's real dimensions. OpenCV solves for the 3×3 homography matrix that maps those four image points onto their real world equivalents, and every subsequent tracked pixel position for the rest of the video is passed through that same matrix to produce a position in metres (Figure 5). Because the matrix is solved once per video rather than re estimated every frame, the whole approach depends on the camera itself staying reasonably still relative to the pitch, panning and zooming are handled separately by the optical flow stage that runs before this one (Stage 8), not by re solving the homography.

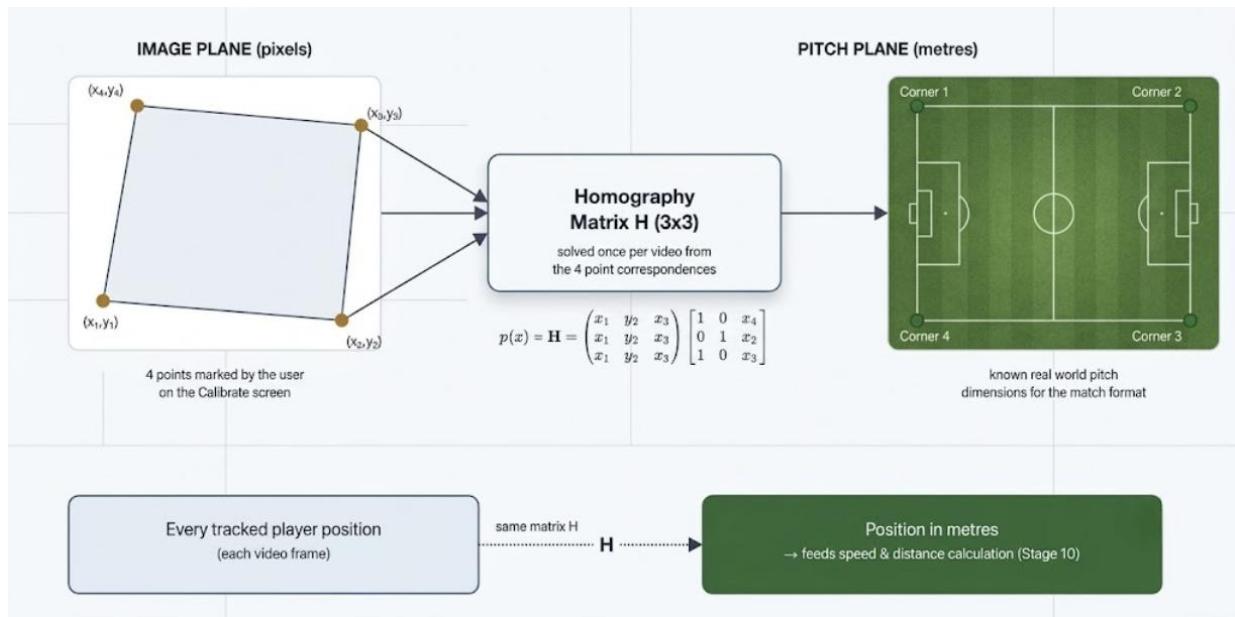


Figure 5 Four point homography: how pixel positions on the Calibrate screen become pitch coordinates in metres.

How team classification works

Team assignment (Stage 5) works in a similarly direct way. Once a player is detected and tracked, a small crop around their torso is sampled and reduced to its dominant colour. K Means clustering, with k fixed at two because football always has exactly two teams, groups every torso colour observed across the whole video into two clusters, no manual "team A wears red" labelling step is required at any point. The referee and goalkeeper detection classes are deliberately excluded from this clustering step, since including them would pull the two cluster centres toward colours that are not actually either team's kit, and the effect would only get worse on the (regularly encountered) matches where a goalkeeper's kit accidentally resembles one team's outfield colour.

A typical request end to end

Figure 6 traces a single video through the whole system, from the moment it is uploaded to the moment its results render on the Analysis dashboard, showing how the frontend, backend,

processing manager, pipeline, and file storage described above actually interact over the course of one job.

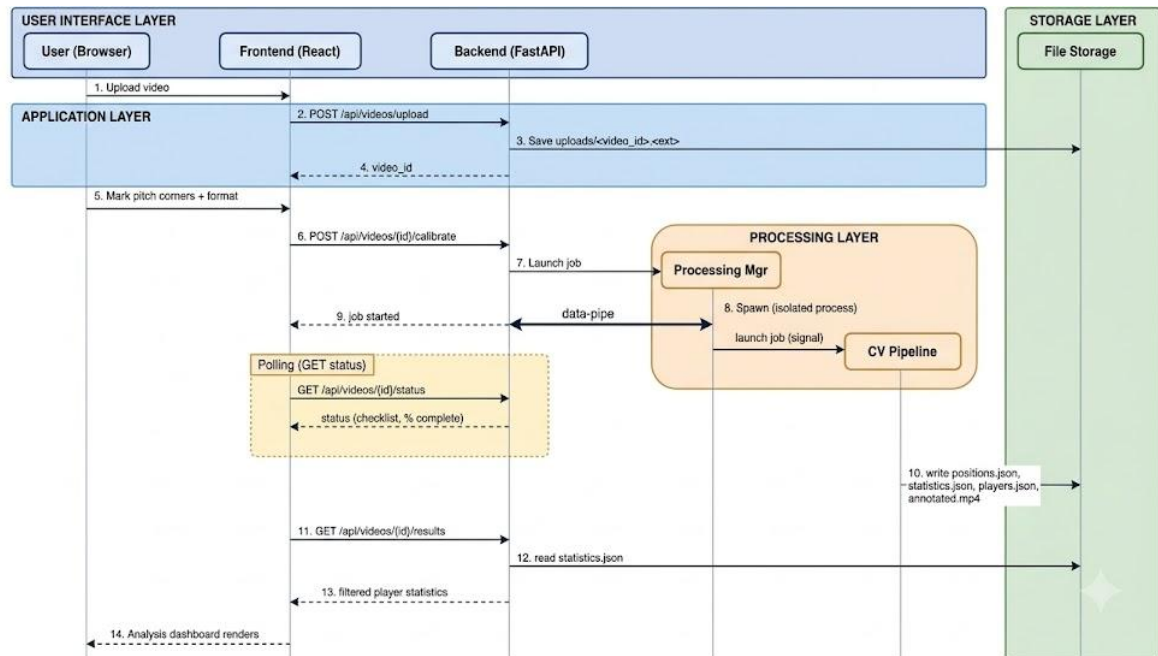


Figure 6 Upload → calibrate → process → results, traced across the system.

4.3 Data model

PitchTrack deliberately uses no database. Every video and its results are addressed by a randomly generated identifier and stored entirely as files on disk, a scope decision recorded in the project's own requirements document, to be revisited only if a defined scale requirement makes file based storage genuinely insufficient, not introduced pre emp

Table 4 File based data model.

Path	Contents
uploads/<video_id>.<ext>	The original uploaded video, unmodified.
results/<video_id>/job.json	Processing status (Uploaded / Processing / Completed / Failed) and the step checklist shown on the Processing screen.
results/<video_id>/statistics.json	Per player average/minimum/maximum speed, total distance, team, match metadata, ball statistics, and team possession percentages.
results/<video_id>/positions.json	Per player, per frame pixel or metre coordinates, used for the raw data endpoint and future overlay work.
results/<video_id>/players.json	The flat list of tracked players and their team assignment.
results/<video_id>/annotated.mp4	The output video with tracking overlays, live per player speed/distance labels, and the possession banner baked in.

4.4 API Design

Instead of exposing a generic CRUD interface for the backend REST API, a small, purpose built API is exposed based on the linear workflow of the dashboard from uploading the files to exporting the analyses.

Table 5 Endpoints of REST API

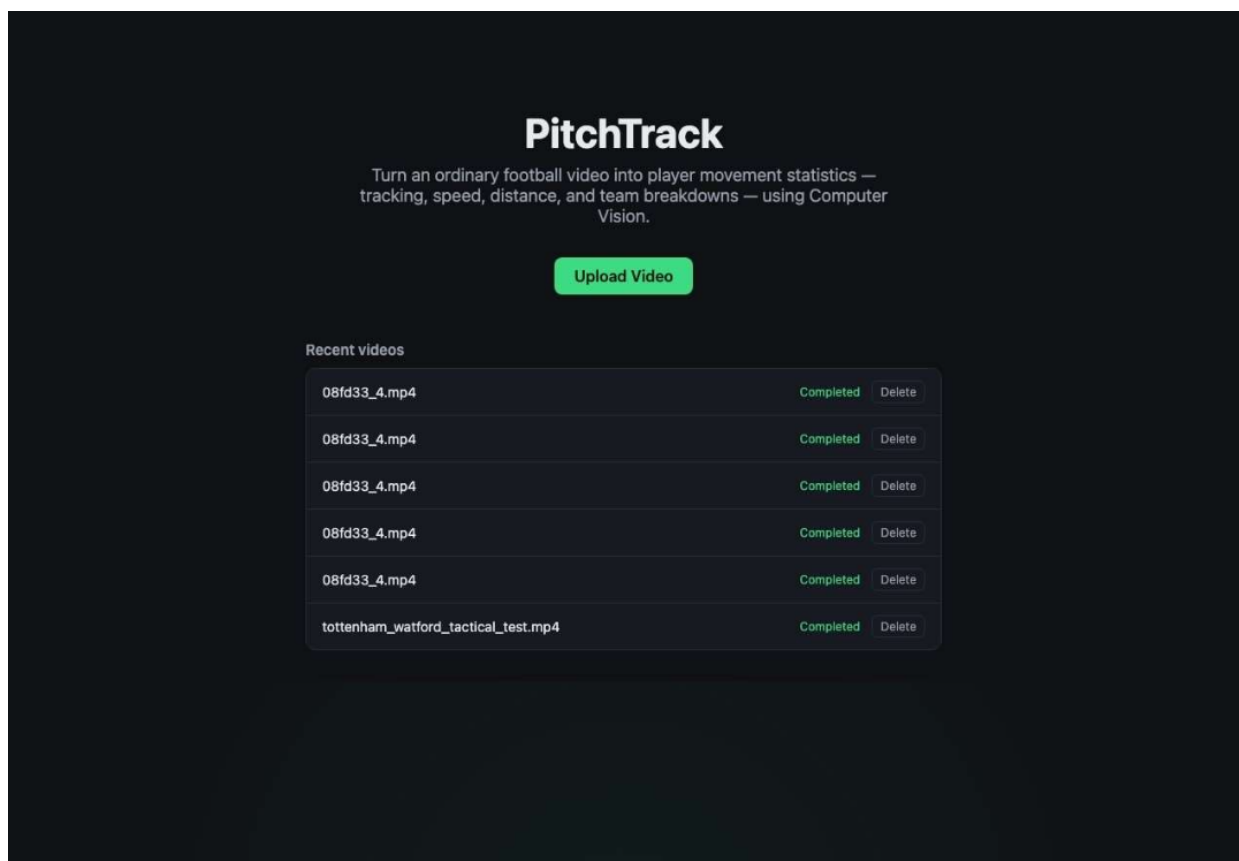
Method & Path	Purpose	Notes
POST /api/videos/upload	Validate and store an uploaded video.	Does not start processing, validates extension, size, and readability only.
POST /api/videos/{id}/calibrate	Submit (or skip) four pitch reference corners and the match format; starts the pipeline.	This call, not upload, is what actually launches the background CV job.
GET /api/videos/{id}/status	Poll job status and the per step checklist.	Used by the Processing screen.
GET /api/videos	List all processed videos.	The file based stand in for a videos table, used by the Home screen's recent videos

		list.
GET <code>/api/videos/{id}/results</code>	Filtered player statistics.	Supports <code>team</code> , <code>tracking_id</code> , <code>start_time</code> , and <code>end_time</code> query filters.
GET <code>/api/videos/{id}/positions</code>	Raw per frame position data.	Metres if calibrated, pixels otherwise.
GET <code>/api/videos/{id}/video</code> , <code>/annotated</code>	Stream the original / annotated video.	Supports HTTP range requests so the browser can scrub without downloading the whole file.
GET <code>/api/videos/{id}/export/{csv json annotated video}</code>	Download results in the requested format.	Honours the same filters as the results endpoint.
DELETE <code>/api/videos/{id}</code>	Cascade delete a video and all its results.	Removes the upload, all JSON results, and the annotated video.

4.5 Dashboard Screens

The dashboard leads users through six screens which appear in a predefined sequence starting with Home (recent videos). The Upload screen enables users to drag and drop files while showing them their upload progress. Users can select their four pitch corners through the Calibrate screen which provides an exact frame timeline and includes the match format selection tool that Section 4.6 explains. The Processing screen shows users their current status through a polling checklist system. The main dashboard exists as the Analysis screen. The system enables users to export their data through

Table 6 Home screen, showing the recent videos list



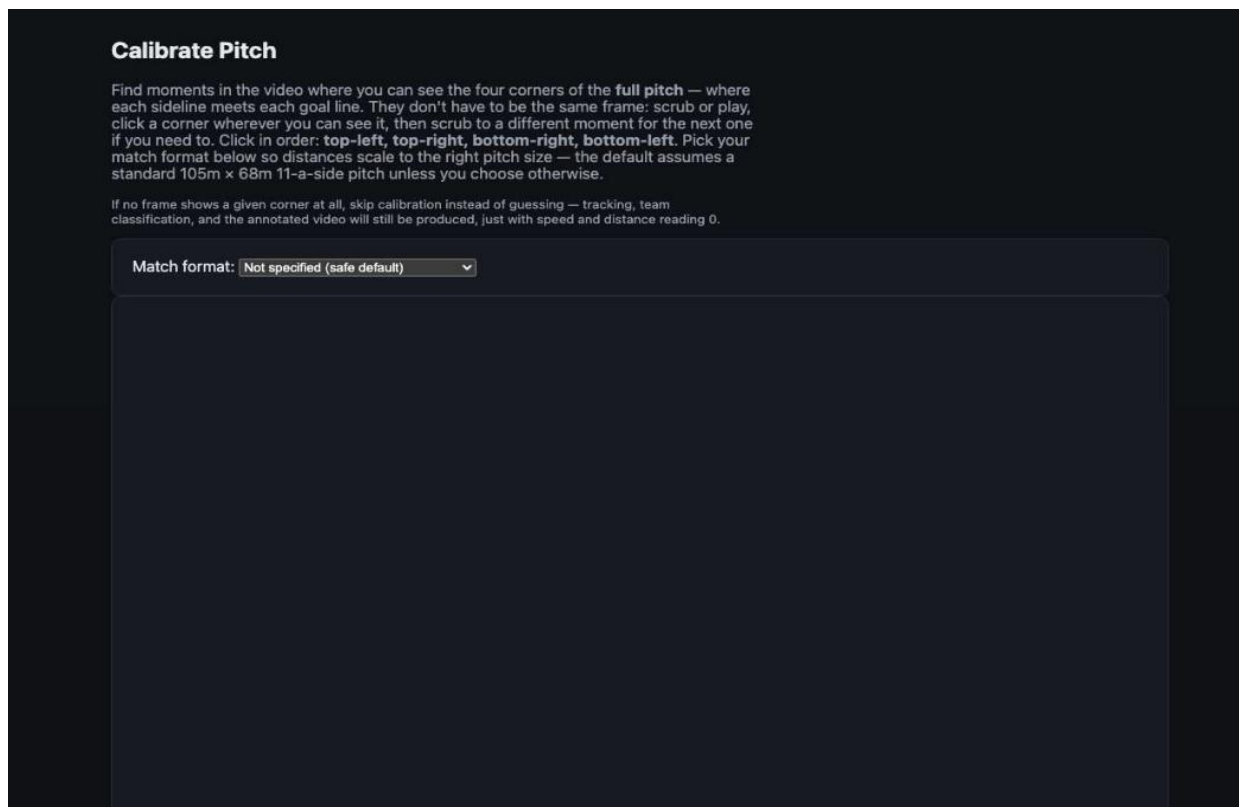


Figure 7 Calibration screen with the match format selector

The Analysis screen (Figure 9) is the system's main deliverable from a user's point of view: a match at a glance KPI strip (players tracked, match duration, top speed, distance leader, possession lead) computed client side from the same results payload used everywhere else, the annotated video, a live possession bar and ball tracking panel, and a sortable per player statistics table with a bar chart comparison of average and maximum speed by player.

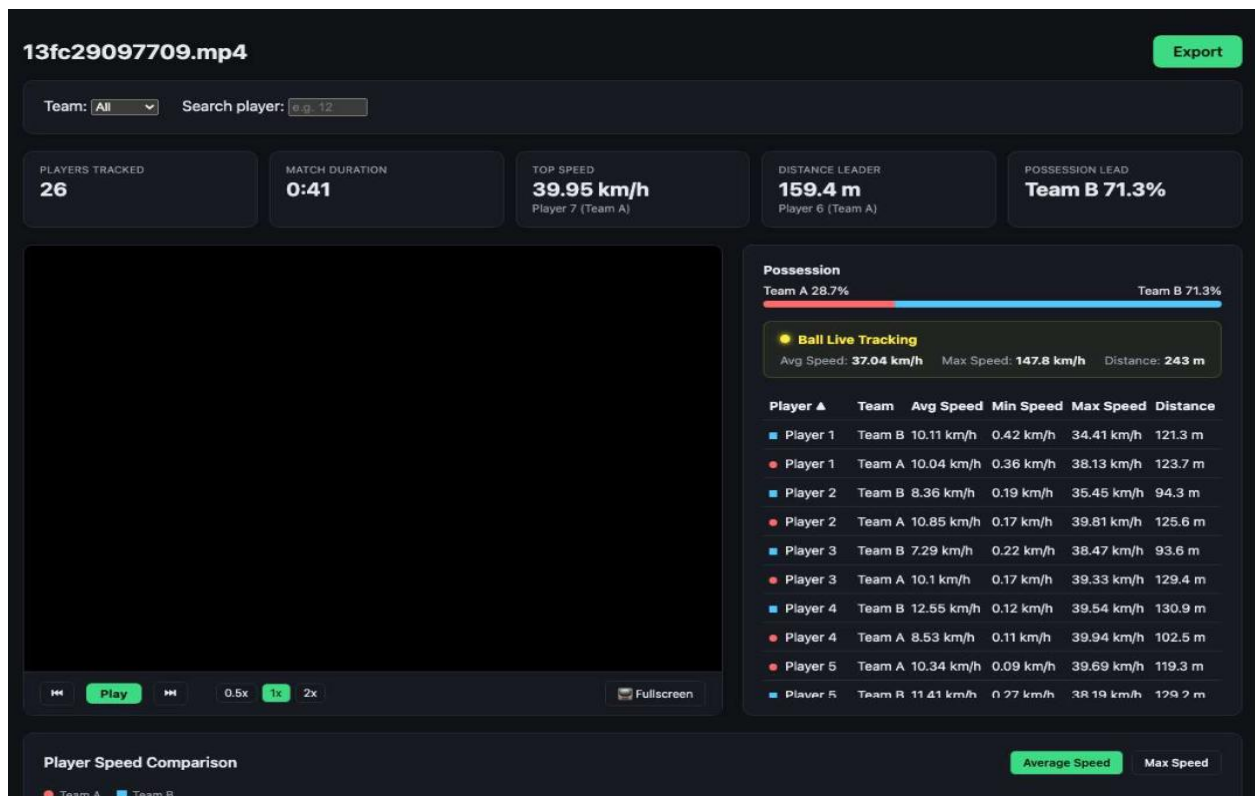


Figure 8 Analysis dashboard: KPI strip, video, possession, and player statistics.

Figure 10 shows a frame from the annotated output video itself, the artefact the whole pipeline exists to produce. Each tracked player appears with a team coloured marker and a team relative ID, the referee is shown in a third, visually distinct colour so as never to be confused with either team, and live speed and distance readings are drawn above each player's head, updating in real time as the video plays.

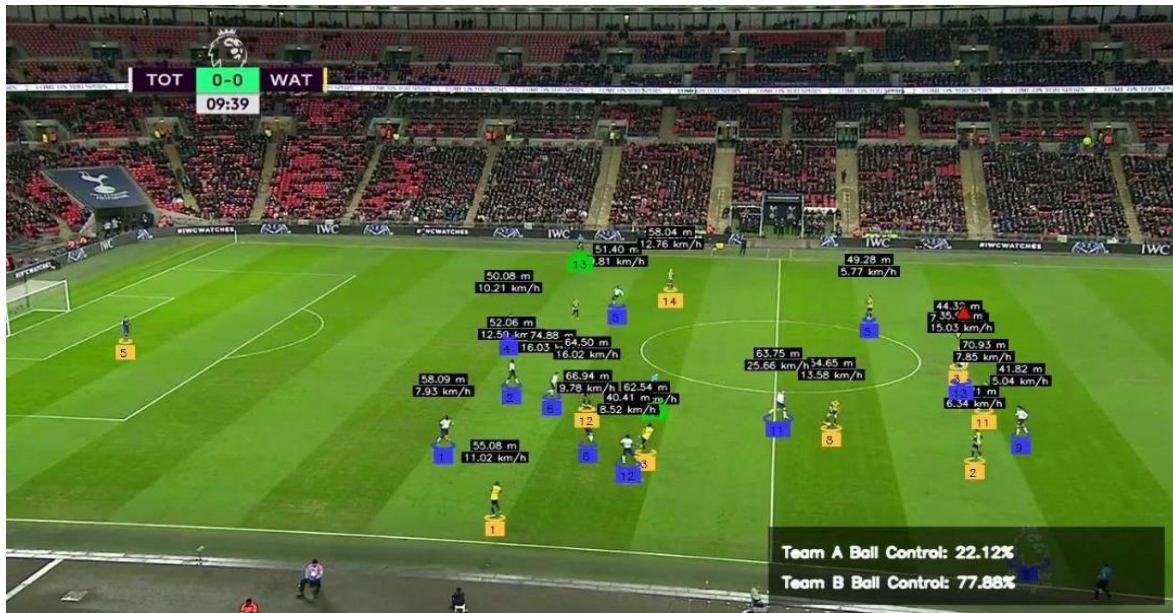


Figure 9 A frame from a real annotated output video, showing player tracking, team separation, referee exclusion, live speed/distance labels, and the ball possession banner.

4.6 Innovation carried by you

Beyond wiring together an off the shelf detector and tracker, this project's own engineering contribution was making the resulting system correct under conditions that a first working version did not handle. Each item below is a specific, verified fix, verified by re running the pipeline against real footage and inspecting the actual output, not only by reading the code, rather than a general description of the technology used.

4.6.1 Configurable match format framework

The tracking stage originally enforced a hardcoded assumption that every match had exactly seven players per team, silently merging or discarding any additional real, correctly detected player once that assumption did not hold, the direct cause of an 11 a side match being mis reported with roughly a third of its real players missing. This was replaced with a configurable match format system (5v5, 7v7, 9v9, 11v11, or a custom player count and pitch size), selected once at the calibration step and threaded through the API into the tracking stage, together with a generous safety margin default, rather than a forced guess, for the case where a user does not specify a format at all.

4.6.2 Referee/player detection suppression fix

By directly comparing two annotated output frames of the same match moment, a real player standing next to the referee was found to have no detection at all in one version of the pipeline, while the referee did. The root cause was a class agnostic non maximum suppression setting: it had originally been enabled to solve a real, different problem, the same person being double detected as both "player" and "goalkeeper", but it also suppressed a genuinely different person, the player, whenever their box overlapped the referee's closely enough. The fix switched to per class suppression plus a second, narrowly scoped pass that still merges only player/goalkeeper duplicates on the same body, confirmed by re rendering the same clip and visually verifying that the previously missing player now appears. This mirrors the detector tuning difficulty Vandeghen et al. (2022) and Golovkin et al. (2025) each report while building their own single camera soccer pipelines against SoccerNet.

4.6.3 Perspective Transform validation bug

While adding support for smaller pitch sizes, needed for 5 a side and 7 a side formats, the component responsible for converting pixel coordinates into real world metres was found to validate its own output against a hardcoded full size 11 a side pitch boundary, regardless of what pitch size it had actually been configured with, meaning a smaller, custom pitch's genuinely valid points could be silently rejected, or a full size pitch's invalid points silently accepted. This was fixed so the check always uses the instance's own configured dimensions, and a regression test was added specifically to catch this class of bug returning.

4.6.4 Speed measurement noise handling

The original speed calculation capped any implausible reading, caused by tracking or calibration noise, at a fixed ceiling value rather than discarding it, which meant a noisy reading

and a genuinely fast, near record sprint were reported as the identical number, and a player's "maximum speed" of exactly the ceiling value gave no way to tell whether that figure was real. The fix excludes implausible readings through complete removal from the calculation instead of clamping them. The system shows its maximum speed through readings which it accepts as valid information. This is The video versus GPS validity research applies its own honesty over fabrication principle which matches the same approach (Crang et al., 2025; Kavanagh et al., 2026): they report a limited and dependable number instead of an unlimited and uncheckable value.

4.6.5 Graceful degradation for partially calibrated tracks

A player whose tracked position never once fell inside the calibrated pitch boundary, in practice, a sign of imprecise corner picking during calibration rather than a real absence of that player, was previously dropped from the results entirely. On one real test this collapsed a roster of over twenty tracked players down to a single visible row, with no indication to the user that anything had gone wrong. The fix makes such a player degrade the same way an entirely uncalibrated video already does, reporting zero speed and distance rather than disappearing, and a diagnostic log line was added for when this affects an unusually large share of a team, since that specifically signals a calibration problem worth revisiting.

4.6.6 Dataset augmentation and measured retraining

The originally deployed model was trained on 372 annotated images. A second, independently sourced dataset of 663 correctly annotated images, using the identical four class label scheme, was located and verified to contain real, non empty annotations, a first candidate dataset was discovered to be an "images only" export with no annotations at all, and was rejected, merged with the original set into a combined 1,035 image training set, and used to retrain the

detection model under the same training recipe as the original model, changing only the data. The resulting per class accuracy change was measured, not assumed (Chapter 5, Figure 12), and the retrained model was deployed only after the improvement had been confirmed.

4.6.7 Independent, out of distribution validation

The training dataset consists exclusively of Bundesliga broadcast footage. To test genuine generalisation rather than performance on data resembling the training distribution, the deployed model was additionally run, after explicit confirmation that the source video was appropriately licensed for this evaluation use, against a Premier League broadcast clip from a different stadium, league, and broadcaster that the model had never seen in training. It correctly recovered the true 11 versus 11 roster count, direct evidence that the fixes described above. The system learns to recognize different video content types through its training on development footage.

5. TESTING & IMPLEMENTATION

5.1 Testing Plan

The testing process consists of three stages which show how a computer vision pipeline operates through its different phases. The system contains standard program logic which developers can fully test through unit testing and it performs actual detection and tracking operations. The system requires evaluation through actual footage because unit tests fail to assess this aspect independently. Figure 11 shows the three phases The system operates through its scheduled sequence of operations which Table 6 explains in complete detail.

Table 7 Testing stages

Stage	What it covers	Examples
Unit testing	The system tracks team merging activities and roster cap choices while it applies perspective transformation limits and manages speed and distance-based exclusion patterns. The system groups teams into separate categories through its team classification process. The system applies detection suppression techniques which work differently for each class. The system determines the correct match format through its resolution process.	test_tracking_merge.py, test_perspective_transform.py, test_speed_distance.py, test_team_classification.py, test_detection_nms.py
Integration testing	The API surface end to end against a mocked pipeline: upload validation, the calibrate then process handoff, range request video streaming, and result filtering.	test_upload_and_calibrate.py, test_range_response.py, test_filtering.py
Real footage validation	The trained model needs to run on actual broadcast video content because unit tests fail to prove its detection and tracking abilities. The trained model needs to run on actual broadcast video content because unit tests fail to prove its detection and tracking abilities. The trained model needs to run on actual broadcast video content because unit	Manual verification against a Bundesliga sample clip and a separate Premier League clip never used in

		training
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The automated suite contains 53 passing tests which run under pytest for unit and integration testing and automated test execution. The testing process includes multiple tests which focus on discovering actual regression faults that appeared during the development phase. The testing process included a specific test which verified that the roster cap default setting should not automatically switch back to its previous hardcoded value. The development team created this test during the initial defect resolution process which they documented in Section 4.6.1.

5.2 Implementation

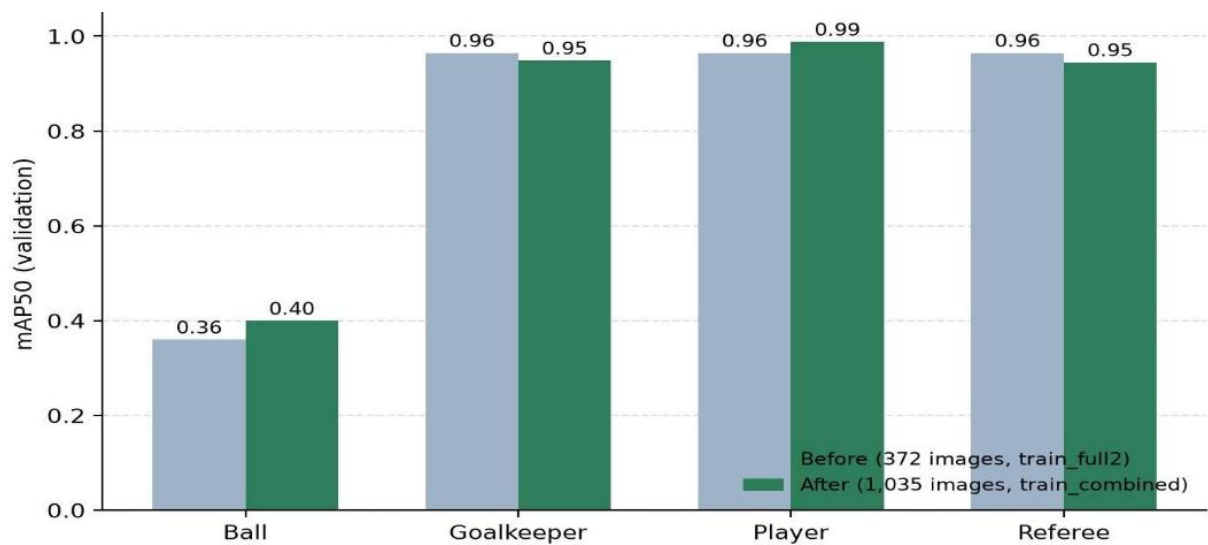


Table 8 illustrates the accuracy of PitchTrack in relation to the accuracy figures presented in the literature for similar systems.

Class	mAP50 before (372 img.)	mAP50 after (1,035 img.)	Comparable literature figure
Player	0.96	0.99	≈ 0.95 to 0.99 across the reviewed literature's people detection tasks (e.g. Aharon et al., 2022: 80.5 MOTA on MOT17)
Goalkeeper	0.96	0.95	Comparable strong class detection; no goalkeeper specific benchmark identified in the literature reviewed
Referee	0.96	0.95	Comparable strong class detection; Golovkin et al. (2025) and Tshiani (2026) both report high match official precision
Ball	0.36	0.40	Vicente-Martínez et al. (2023) report the same accuracy ceiling on soccer ball detection specifically; Vandeghen et al. (2022) report 52.3% on the mixed class SoccerNet v3 benchmark

From the information contained within Table 7, it is clear that detecting persons in the frame is a well-solved problem compared to detecting the ball itself. This finding is consistent across all of the systems reviewed in the literature, and is replicated in PitchTrack’s measure performance statistics.

The backend of the system runs as a Uvicorn ASGI process, while the frontend is served by a Vite development server. Communication between the two is through a locally scoped REST

API. The trained detection model is deployed as a single weights file (app/pipeline/models/best.pt); every time a retrained model was deployed during this project, the previous weights file was backed up first, so a regression could be reverted with a single file copy rather than a retrain.

Version control was itself a concrete implementation task rather than an assumed given. The project's history was initially entangled inside a larger, multi project personal repository that also contained unrelated coursework. Before publishing the project, its real commit history was extracted into its own independent repository, preserving the actual sequence of development commits rather than discarding it for a single fresh "initial commit", and published to a dedicated GitHub repository with a single default branch and a clean file listing, excluding large training data folders, generated results, and any copyrighted footage used only for local testing.

6. SUMMARY

Building PitchTrack end to end has been a substantial exercise in applied computer vision and full stack engineering, but the more durable lesson from this project is methodological: a computer vision pipeline that runs without crashing is not the same thing as a pipeline that produces correct output, and the gap between the two is only found by deliberately trying to break the system against real, messy data and tracing what actually goes wrong back to its cause. A crashing native library conflict, a badly biased colour clustering heuristic, a detection suppression setting that fixed one bug while quietly causing another, a calibration edge case that could silently erase most of a match roster, none of these were found by reading the code carefully enough in advance; all of them were found by running it and looking closely at what came out. The project team discovered real value from keeping an automated test suite which operated as their official testing system. The practice of maintaining automated tests throughout this project proved its worth through actual results which went beyond standard theoretical benefits. The development process revealed a regression which caused a detection class to disappear after its initial resolution. The existing test system detected the problem right away which prevented it from developing into a hidden data quality issue that would have affected the production environment. The research team gathered additional information for model training while they assessed object class accuracy changes through direct measurement instead of estimation.

The research showed that extra data collection did not create equal accuracy improvement for all object classes because it brought small changes to the ball class yet maintained its existing performance for other classes. The process of creating an expanded version taught me that individual design decisions become more credible when experts use their current knowledge to evaluate them. Every pipeline stage discussed in Chapter 2 turned out to have an active body of research behind it, sports specific multi object tracking (Huang et al., 2022; Cao & Li, 2026;

Adžemović, 2025), occlusion recovery limits for appearance based re identification (Ranasinghe & Thayasivam, 2025), automatic sports field calibration (Theiner & Ewerth, 2022; Gutiérrez-Pérez & Agudo, 2024), and the continued difficulty of small, fast ball detection as recently as 2024 (Raj et al., 2024), none of which was consulted while the original engineering decisions in Chapter 4 were actually being made, yet all of which turned out to support those decisions rather than contradict them. That agreement is itself a useful, if retrospective, check on the project's design.

6.1 Limitations

- Ball detection remains the weakest class (validation mAP50 $\approx$ 0.40 versus approximately 0.95 to 0.99 for player, goalkeeper, and referee), because a football is a small, fast moving, frequently occluded object, a known, literature confirmed limitation of this class of single camera detection (Section 2.6), not unique to this implementation.
- Roster size correction is a statistical safety net tuned against the footage available during development, not a guarantee, particularly on matches with heavy, prolonged player occlusion, consistent with the 50% heavy occlusion recovery rate and 8.66% long absence re acquisition rate reported for a comparable appearance based football tracker (Ranasinghe & Thayasivam, 2025).
- Pitch calibration accuracy depends on the user manually and precisely marking four pitch corners; imprecise corner placement propagates directly into speed and distance error, in the same way camera and correspondence quality bound accuracy for the commercial systems studied by Crang et al. (2025) and Kavanagh et al. (2026).
- There is no database and no user accounts, by deliberate scope decision, the system

does not currently support multiple concurrent users managing a shared video library with access control.

- The pipeline assumes a single, relatively stable elevated broadcast or tactical camera angle; it has not been validated against a low, pitch level, handheld, or rapidly panning camera.
- Processing is offline and batch based on a pre recorded video, not real time or live streamed.

6.2 Future Scope

- Collecting and fine tuning specifically on additional ball only imagery, or exploring a dedicated small object detection technique such as a heatmap based, motion attention architecture, to close the remaining gap in ball detection accuracy identified in Chapter 5 (Huang et al., 2019; Raj et al., 2024).
- Automatic pitch line and keypoint detection, to remove manual four corner calibration entirely and replace a user driven step with a fully automated one, an approach already demonstrated in the sports field registration literature reviewed in Section 2.8 (Theiner & Ewerth, 2022; Gutiérrez-Pérez & Agudo, 2024), rather than a purely speculative direction.
- Live or streaming input support, extending the current offline/batch design.
- The implementation of a suitable database system begins when file storage fails to meet the needs of multiple users and tracking historical data becomes necessary. The decision to move away from file-based storage systems was made because these systems no longer meet the requirements of projects which have already defined their needs. The project requirements already contain the decision which determines the transition from file storage to database systems.
- The current project phase does not include the development of a mobile friendly frontend design which remains unaddressed.
- The detection of tactical events together with passes and shots and goals and offside situations exists as a separate and much larger upcoming project which we established from the beginning would not be part of this current project. The research area about tactical event detection and passes and shots and goals and offside situations exists as an

independent field which developed its own body of literature through the work of Zhao and his colleagues in 2023. The research group studies two problems which exist independently from each other because they require separate solutions.

- The project will replace its current K Means team classification system with a learned contrastive embedding method which handles teams that wear similar white or dark uniforms based on the successful approach by Koshkina et al. (2021) after the present native library dependency restriction (Section 3.1) which caused the use of a simple classifier becomes irrelevant.

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List of Symbols, Abbreviations and Nomenclature

CV — Computer Vision

YOLO — You Only Look Once (a real-time object-detection model family)

NMS — Non-Maximum Suppression

Re-ID — Re-Identification (appearance-based identity matching)

K-Means — a clustering algorithm used here for jersey-colour team classification

mAP50 — mean Average Precision at an Intersection-over-Union threshold of 0.50

FPS — Frames Per Second

API — Application Programming Interface

REST — Representational State Transfer

JSON — JavaScript Object Notation

KPI — Key Performance Indicator

ASGI — Asynchronous Server Gateway Interface

MPS — Metal Performance Shaders (Apple Silicon GPU acceleration backend)

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